

Deep learning based analysis of spontaneous speech for diagnostic classification and biomarker prediction in Alzheimer’s disease and primary progressive aphasia

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Abstract

Identifying underlying amyloid-beta ($A\beta$) pathology, a hallmark of Alzheimer’s disease (AD), typically requires invasive or costly procedures. In addition, differential diagnosis between Alzheimer’s disease (AD) phenotypes and Primary Progressive Aphasia (PPA) variants remains a significant clinical challenge due to overlapping symptomatic profiles. This work evaluates the utility of a deep learning-based spontaneous speech analysis for biomarker prediction in AD and diagnostic classification in PPA. While existing models are biased toward English-language datasets, we benchmark our framework on the IS2021 ADReSSo challenge and extend it to a new Spanish-speaking cohort from the Sant Pau Initiative on Neurodegeneration (SPIN). Our architecture leverages a deep learning system that integrates state-of-the-art encoders for AD detection and audio processing. Our approach integrates acoustic and linguistic features for the automated clinical phenotyping of neurodegenerative diseases.

Index Terms: Alzheimer’s disease detection, primary progressive aphasia, amyloid-beta prediction, spontaneous speech analysis, multimodal deep learning, speech biomarkers

1. Introduction

Amyloid-beta ($A\beta$) accumulation is a hallmark pathological biomarker of AD [1], appearing decades before clinical symptoms manifest. However, traditional diagnostic methods, such as cerebrospinal fluid analysis or Positron Emission Tomography (PET) neuroimaging, are invasive, costly, and difficult to scale. These barriers often lead to diagnoses being made years after symptom onset, missing the crucial therapeutic window for early intervention. This delay is particularly concerning given the recent emergence of anti-amyloid disease-modifying therapies, whose clinical efficacy depends on the early identification of $A\beta$ pathology.

While $A\beta$ represents a primary biological target, the diagnosis of neurodegenerative diseases remains a significant clinical challenge due to the high degree of shared symptomatology among different syndromes. Language impairments, in particular, are frequent and often early manifestations of conditions such as Alzheimer’s disease (AD) and frontotemporal dementia (FTD). A primary example of this linguistic overlap is Primary Progressive Aphasia (PPA), a clinical syndrome characterized by a selective and progressive decline in language abilities. PPA is currently classified into three distinct variants based on the predominant speech deficit: the non-fluent/agrammatic variant (nfvPPA), marked by effortful speech and grammatical errors; the semantic variant (svPPA), defined by a loss of word meaning and anomia; and the logopenic variant (lvPPA), characterized by phonological processing deficits and impaired sentence

repetition [2, 3]. These clinical variants serve as windows to different underlying molecular pathologies. While lvPPA is most frequently associated with Alzheimer’s pathology, svPPA and nfvPPA are more commonly linked to the Frontotemporal Lobar Degeneration (FTLD) spectrum [4]. Accurate classification of these variants guides clinicians toward the correct underlying pathology and informs treatment strategies. However, distinguishing between these variants in a clinical setting is difficult due to overlapping symptoms and the high level of expertise required for manual linguistic analysis.

In response to these challenges, automated speech analysis has emerged as a promising, non-invasive digital biomarker [5, 6]. Recent reviews indicate that deep learning models, especially those that integrate acoustic and linguistic features through pre-trained transformers like BERT [7] or wav2vec [8], can detect AD from spontaneous speech with high accuracy [6]. Similarly, the application of Artificial Intelligence (AI) to PPA has shown immense potential for improving diagnostic precision and subtype classification [9]. While spontaneous speech assessment is a proven, high-utility approach for AD and PPA diagnosis, the vast majority of existing literature focuses on English-speaking cohorts, frequently utilizing the benchmark ADReSSo dataset [10]. As a widely recognized standard for AD classification, ADReSSo provides a robust point of technical comparison for the present work. To address the linguistic gap in current research, we utilize the novel SPIN (Spanish Interdisciplinary) cohort [11], a dedicated Spanish-language dataset. The use of both datasets allows this work to establish a language-specific baseline.

The Double Multi-Head Attention Multimodal (DMHAM) system is adapted from high-performing architectures originally designed for speaker recognition [12]. This system has been successfully repurposed for other speech classification tasks, such as speech emotion recognition (SER) [13], where it has demonstrated competitive results in different SER challenges. Specifically, it ended up in the first place in the IberLEF (SEPLN) competition [14], the third place in the Odyssey’s 2024 one [15], and seventh place in Interspeech 2025 [16]. The DMHAM aims to capture speech and text features by making use of a dual stream approach that concatenates and integrates them through an attention-based pooling mechanism. However, other works like CogniAlign [17] introduce a more sophisticated fusion strategy designed to temporally align acoustic and linguistic cues, providing a potentially better representation of the information extracted from the audio.

The primary aim of this work is to address the significant linguistic imbalance in current dementia research by introducing a new Spanish-speaking dataset to establish a performance baseline, as the vast majority of existing literature remains focused on English-speaking populations. We adapt and evaluate

the DMHAM, our high-performing deep learning architecture mentioned above, to the clinically distinct task of $A\beta$ prediction and PPA variant classification, testing its diagnostic efficacy and its ability to generalize across different speech-based domains. Central to the work’s contribution is a rigorous performance comparison between this model and the CogniAlign framework, which employs a novel fusion strategy designed to more effectively align acoustic and linguistic cues. Furthermore, the authors investigate the specific diagnostic utility of hand-crafted acoustic features to determine their impact on overall model results. This analysis allows for an evaluation of whether traditional clinical markers remain competitive with, or complementary to, end-to-end deep learning representations.

To facilitate this evaluation, the authors incorporate a set of handcrafted acoustic features that capture temporal, prosodic, voice-quality, and spectral characteristics of spontaneous speech. Among these features, pause-related measures are of particular interest, as previous studies have shown that they are salient markers in Alzheimer’s disease [17, 18]. In particular, a higher pause rate, longer pauses, and a lower speech-to-pause ratio may reflect difficulties in lexical retrieval, speech planning, and fluency, making them especially informative and clinically interpretable markers. We also consider voice-quality features such as jitter and shimmer, which measure instability in vocal frequency and amplitude and may capture subtle phonatory alterations associated with cognitive decline [19, 20].

The remainder of this paper is structured as follows. Section 2 reviews related work on speech-based detection of $A\beta$, AD and PPA, highlighting recent advances in machine learning and deep learning approaches. Section 3 describes the proposed methodology, including the multimodal architectures and feature integration strategies. Section 4 outlines the experimental setup, datasets, and training procedures. Section 5 presents the results and comparative analysis across tasks. Finally, Section 6 discusses the findings and concludes the study.

2. Related work

Early computational approaches to Alzheimer’s Disease (AD) focused primarily on distinguishing symptomatic patients from healthy controls using linguistic and acoustic features [21]. More recently, researchers have shifted toward the more challenging task of predicting biological pathology, specifically $A\beta$ positivity, which often precedes clinical symptoms.

Much of the foundational work in $A\beta$ prediction utilizes machine learning (ML) models, such as Random Forest and Support Vector Machine. Previous research demonstrated the efficacy of speech-based screening for $A\beta$ [22]. Similarly, research involving cognitively unimpaired Spanish-speaking cohorts [23] indicates that although speech may serve as a non-invasive proxy for $A\beta$, its predictive value is limited before overt symptoms emerge. In the context of PPA, connected speech markers have been shown to identify amyloid burden with notable classification accuracy [24]. Recent analyses on spontaneous speech in AD patients and healthy controls found that traditional Random Forest models demonstrated moderate sensitivity, and conventional acoustic features yielded only modest classification accuracies [5]. These results underscore that, while the acoustic and linguistic markers are predictive, there is a performance gap in traditional predictive models.

To bridge this divide, the field is evolving beyond classical machine learning toward end-to-end deep learning architectures that can capture more complex speech patterns. This evolution suggests that multi-pipeline systems, which combine self-

supervised learning (SSL) audio processing with high-fidelity transcriptions, may be most promising for identifying the subtle patterns associated with $A\beta$ pathology. Central to this shift are state-of-the-art SSL frameworks such as wav2vec 2.0 [25], HuBERT [26], and WavLM [27]. All of them employ transformer-based architectures to encode raw audio files. A review of speech-based detection of AD based on AI techniques reveals the dominance of deep learning models and the integration of acoustic and linguistic features [6].

The differential diagnosis of PPA variants (nfvPPA, svPPA and lvPPA) remains a significant clinical challenge due to overlapping symptomatic profiles. High-dimensional linguistic and acoustic analysis has shown remarkable potential in PPA multi-class separation. Rezaii et al. [28] achieved a state-of-the-art accuracy of 97.9% using generative AI large language models (LLMs) and hand-crafted linguistic features. A combination of lexical and acoustic markers yielded robust classification results in discriminating patients from controls and in the differential diagnosis of svPPA and nfvPPA [29]. However, these approaches rely on labor-intensive handcrafting of speech features. Recent advances in AI have sought to move from manual feature engineering to end-to-end deep learning architectures. Deep Neural Networks (DNNs) that integrated acoustic and linguistic features extracted automatically from speech achieved classification accuracies of 80% for the three PPA variants, outperforming both classical models and the diagnostic benchmarks of expert clinicians [30].

A growing body of literature highlights specific acoustic “fingerprints” that differentiate the variants. Recent analyses identified six critical language measures that distinguish patient groups, including pause rate, fillers, and lexical diversity [31]. Specific acoustic features provide even higher resolution; for instance, articulation rate has demonstrated the capacity to differentiate nfvPPA from svPPA with high precision [32]. Building on this, a more sophisticated temporal metric (Narrative Syllable Duration, NSD) was found to differentiate nfvPPA from lvPPA with near-perfect accuracy [33]. Fundamental frequency (f_0) range and pause played a crucial role in distinguishing nfvPPA from svPPA [34]. These findings validate the inclusion of dedicated acoustic pipelines in deep learning models to capture the rhythmic and prosodic deviations characteristic of each variant.

A critical gap remains in applying these models to non-English datasets, where linguistic structures differ. The present work study addresses this by focusing on Spanish-language speech. Previous research has demonstrated that ML classifiers optimized for Spanish linguistic features can achieve exceptionally high diagnostic accuracy [35]. Recent work [36] on Catalan-Spanish bilinguals underscores this complexity, finding that acoustic-based classifiers showed robust performance even in a patient’s non-dominant language, while linguistic-based models tended to demonstrate comparatively lower predictive power.

3. Methodology

This study proposes a multimodal deep learning framework designed to analyze spontaneous speech for the detection of neurodegenerative diseases. The system processes both acoustic and linguistic information extracted from speech samples. The core methodology involves extracting high-level representations from raw audio and automatic transcriptions using state-of-the-art pre-trained encoders, followed by a fusion mechanism that integrates these modalities to perform diagnostic clas-

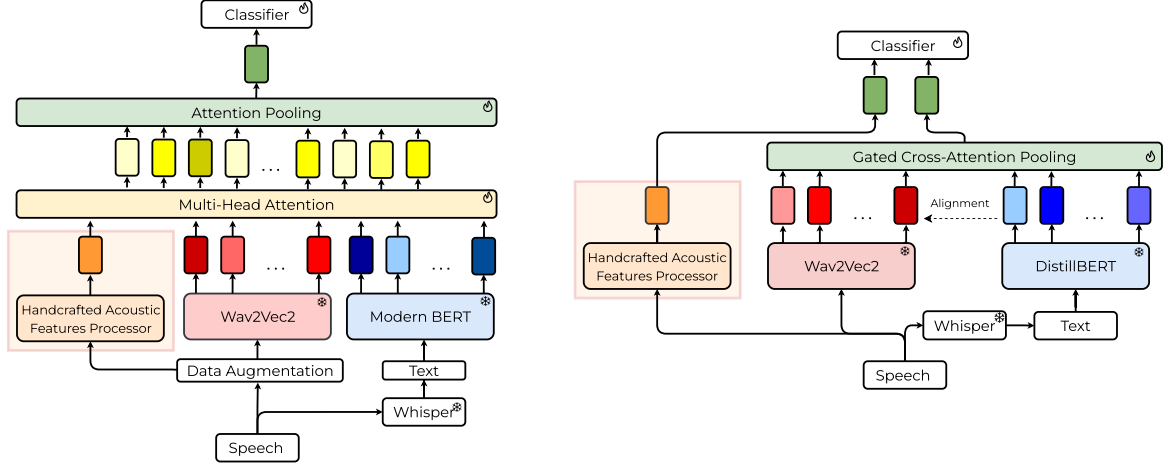


Figure 1: Architectural comparison of the two multimodal frameworks evaluated in this study. The **left** panel illustrates the Double Multi-Head Attention Multimodal (DMHAM) system. The **right** panel depicts the CogniAlign framework.

sification. We investigate two distinct architectures for these tasks: the Double Multi-Head Attention Multimodal System (DMHAM), and a state-of-the-art comparative model, CogniAlign. Both models are designed to handle the variable length of spontaneous speech and leverage transfer learning from large-scale pre-trained models. In the following subsections, we first describe each core architecture independently, and then present the strategy by which both are augmented with handcrafted acoustic features.

3.1. DMHAM (Double Multi-Head Attention Multimodal System)

The DMHAM architecture utilized in this work, shown in Fig. 1, is based on the framework established in [15], and the subsequent advancements introduced in [13], which we seek to validate on the novel SPIN cohort. Adapted from recent advancements in speech emotion recognition and paralinguistics, it employs a dual-stream approach:

- **Encoders:** For the acoustic modality, the authors utilize **Wav2Vec2-XLSR-300M** [37] to extract frame-level representations. This model features 300 million parameters and was pre-trained on 436,000 hours across 128 languages, offering strong cross-lingual robustness. To further enhance model generalization and mitigate overfitting on limited clinical cohorts, the acoustic stream incorporates speech data augmentation techniques during training. For linguistic features, the authors employ **ModernBERT-base** [38], a modernized encoder-only transformer comprising 149 million parameters. Trained on a massive corpus of approximately 2 trillion tokens, this architecture features a native context length of 8192 tokens and incorporates rotary positional embeddings, GeGLU activations, and alternating local-global attention.
- **Multi-Head Attention & Classification:** Features from both modalities are projected to a common dimension via linear adapters and concatenated along the feature axis. A Multi-Head Self-Attention (MHA) mechanism is then applied to the multimodal sequence. Subsequently, an attention pooling mechanism aggregates the temporal dimension by learning to weight each timestep, yielding a fixed-size representation

that is passed to a Multi-Layer Perceptron (MLP) classifier.

3.2. CogniAlign Framework

We also evaluate the CogniAlign architecture, Fig. 1, recently proposed by Ortiz-Perez et al. [17], which introduces a more sophisticated fusion strategy to better align acoustic and linguistic cues:

- **Encoders:** This architecture employs **Wav2Vec2-base-960h** for acoustic modeling. For the textual component, it utilizes **DistilBERT**, a lighter and faster transformer alternative that maintains highly competitive linguistic representation capabilities.
- **Word-Level Temporal Alignment:** Unlike DMHAM, which relies on simple concatenation, CogniAlign explicitly bridges the sequence-length gap between modalities. It maps the high-frequency acoustic frames to their corresponding textual tokens, synchronizing the modalities into a cohesive, word-by-word sequence.
- **Gated Cross-Attention & Classification:** Operating on these synchronized sequences, CogniAlign fuses the modalities using a Gated Cross-Attention mechanism. One modality serves as the *query* while the other acts as the *key* and *value*, enabling dynamic cross-modal weighting. A gating mechanism further refines this integration to suppress noise. The fused representation is then aggregated via mean pooling and passed to a final classification head.

Crucially, as a direct result of the *word-level temporal alignment* described above, the model intrinsically captures temporal speech dynamics. By structurally synchronizing the modalities at the word level during the cross-attention phase, vital information regarding speech rate, silences, and word-finding pauses is naturally encoded directly within the multimodal embeddings, well before any explicit acoustic feature engineering is applied.

3.3. Integration of Handcrafted Acoustic Features

Having established the two core architectures, we now describe the strategy by which both are enriched with a complementary set of handcrafted acoustic features. These 55 features are designed to capture fine-grained paralinguistic cues that may be

indicative of cognitive decline but are not easily extracted by SSL encoders operating on the raw waveform alone. The feature extraction methodology is adapted from Ntampakis et al. [39], whose NeuroXVocal framework demonstrated the effectiveness of this feature set on the ADReSSo benchmark. Our implementation follows their pipeline and extends it to the SPIN cohort. Features are extracted using librosa and Praat-based analysis (via parselmouth), and encompass:

- **Temporal and Pause Features:** Total speech duration, speech-to-pause ratio, number of pauses, and statistics (average, maximum, standard deviation) of pause durations. These capture hesitation patterns and fluency disruptions commonly observed in neurodegenerative conditions.
- **Acoustic-Prosodic Features:** Pitch statistics (mean, standard deviation, range), intensity measures, speaking rate, and articulation rate, reflecting the melodic and rhythmic aspects of speech production.
- **Voice Quality Features:** Local jitter, shimmer, and harmonics-to-noise ratio (HNR), capturing voice stability and phonation quality.
- **Spectral Features:** Spectral centroid statistics, zero-crossing rate, and formant frequencies (F1, F2, F3 with mean and standard deviation), characterizing the spectral envelope and articulatory patterns.
- **Cepstral Features:** The first 13 Mel-Frequency Cepstral Coefficients (MFCCs) with their mean and standard deviation, providing a compact representation of the spectral shape.

All 55 features are Z-score normalized per feature dimension across the training split of each fold to prevent data leakage, and are fused into the two architectures at different stages of processing, as detailed below.

In the DMHAM framework, the acoustic feature vector is injected at the input of the attention pooling layer, *before* the final temporal aggregation step. Specifically, the 55-dimensional feature vector is concatenated with the sequence of multimodal embeddings produced by the MHA fusion module, allowing the subsequent attention pooling mechanism to weigh the handcrafted features alongside every temporal frame of the SSL representations. This design choice is motivated by the nature of MHA, as presenting the deterministic acoustic features as an additional input token at the pooling stage allows the attention weights to explicitly learn which temporal regions of the sequence are most complementary to the global acoustic cues, enabling a richer and more interpretable aggregation.

In the CogniAlign framework, the handcrafted features follow a *late-fusion* strategy. After the Gated Cross-Attention module has fully aligned the SSL audio and text representations, and after the subsequent transformer encoder layer and mean pooling step have produced a fixed-size embedding, the 55-dimensional acoustic vector is concatenated directly to this pooled embedding immediately before the final classification MLP. This deliberate separation of concerns allows the gated cross-attention mechanism to operate exclusively on the self-supervised representations without interference from deterministic, corpus-level statistics, thereby preserving the model’s capacity to learn subtle cross-modal dependencies. The handcrafted features are then introduced as a complementary, interpretable signal at the decision boundary, where their global, utterance-level nature is most naturally aligned with the aggregated representation.

Table 1: *Demographics by Amyloid Status*

	Negative (N=49)	Positive (N=53)
Age (years)	72.8 ± 6.1	72.6 ± 6.8
Gender:		
Female	25 (51.0%)	22 (41.5%)
Male	24 (49.0%)	31 (58.5%)
Education years	12.8 ± 6.6	11.8 ± 5.7

Values are mean ± SD or n (%).

Table 2: *Demographics of PPA Variants.*

	lvPPA (N=35)	nfvPPA (N=22)	svPPA (N=17)
Age (years)	72.6 ± 6.5	74.7 ± 4.9	73.4 ± 5.8
Gender:			
Female	15 (42.9%)	9 (40.9%)	11 (64.7%)
Male	20 (57.1%)	13 (59.1%)	6 (35.3%)
Education years	10.4 ± 6.4	12.6 ± 5.9	9.8 ± 6.9

Values are mean ± SD or n (%).

4. Experimental Setup

4.1. Datasets

We evaluate both models on two distinct datasets, covering different languages and pathologies.

ADReSSo-IS2021: The Alzheimer’s Dementia Recognition through Spontaneous Speech only (ADReSSo) dataset [10] is a well-established benchmark derived from the Dementia-Bank Pitt Corpus, consisting of 237 picture description recordings (approx. 166 minutes) from English-speaking participants balanced between Alzheimer’s Disease (AD) and healthy control (HC) subjects. We use it to benchmark our models against the state of the art.

SPIN Cohort: The Sant Pau Initiative on Neurodegeneration (SPIN) [11] is a longitudinal cohort for neurodegenerative disease research at Hospital de la Santa Creu i Sant Pau (Barcelona). We curated a subset of the SPIN cohort consisting of spontaneous speech recordings from Spanish speakers, encompassing clinical diagnoses of PPA and confirmed $A\beta$ status. The ‘Picnic Scene’ (Western Aphasia Battery) [40] served as the stimulus for the spontaneous speech task. The dataset includes a variable number of recordings per subject, ranging from 1 to 7 samples, captured during clinical evaluations. As detailed in Tables 1 and 2, there were no significant demographic differences across groups in either task regarding age, gender, or years of education. Wilcoxon rank-sum test and Pearson’s chi-squared test were used for comparing continuous and categorical variables across groups respectively. We define two tasks on this dataset. The **Amyloid-Beta Prediction** task is a binary classification problem aiming to predict amyloid positivity (positive vs. negative), a key fluid biomarker of AD that typically requires costly or invasive procedures to obtain. For this task, participants were categorized by $A\beta$ status (positive, n=53 vs. negative, n=49). The class distribution is approximately balanced (98 negative, 114 positive samples), lending reasonable reliability to the reported results. The **PPA Classification** task is a 3-class problem distinguishing between patients’ groups (lvPPA n=35, nfvPPA n=22, svPPA n=17, with 104, 63, and 35 samples respectively) This task is considerably more challenging due to the marked class imbalance, with svPPA representing fewer than a third of the lvPPA samples. The limited number of

patients per class means that performance estimates are subject to high variance across folds and that results for the PPA task should be interpreted with caution.

4.2. Training Details

All models are trained using the PyTorch framework. For the ADReSSo dataset, we follow the standard train/test split provided by the challenge. For the SPIN dataset, we employ a 5-fold cross-validation scheme stratified by patient ID to ensure no subject overlap between training and validation sets. All recordings are truncated to the first 40 seconds of speech, which approximates the effective input length used by CogniAlign and avoids including clinician speech that sometimes appears at the end of the recordings.

Optimization is performed using AdamW with a learning rate of $1e-4$ for DMHAM and $2e-5$ for CogniAlign, with a weight decay of 0.1. We use a cosine annealing learning rate scheduler with warmup. The batch size is set to 32. For the $A\beta$ prediction task, we reduced the batch size to 16 and the learning rate to $5e-6$ to account for the dataset characteristics. To mitigate class imbalance in the PPA classification task, we employ a WeightedRandomSampler that oversamples minority classes during training.

For the DMHAM baseline, data augmentation is applied during training, including reverberation, additive background noise, and speed perturbation, to improve robustness. CogniAlign focuses on architectural regularization through dropout (0.1-0.3) and layer normalization.

Our experimental evaluation focuses on three primary tasks: AD classification on the ADReSSo benchmark, and $A\beta$ prediction and PPA classification on the Spanish-speaking SPIN cohort. Performance is assessed using accuracy (Acc), macro-averaged F1-score (F1), recall (Rec), and precision (Prec).

It is important to clarify the training paradigm utilized across our experiments. While our framework heavily leverages transfer learning in the form of frozen, pre-trained self-supervised feature extractors (e.g., Wav2Vec 2.0, ModernBERT, DistilBERT) to obtain foundational embeddings, no model weights were transferred or fine-tuned between the distinct clinical datasets or tasks. For each independent classification objective—AD detection on ADReSSo, $A\beta$ prediction on SPIN, and PPA sub-typing on SPIN, all trainable parameters within the attention mechanisms, fusion modules, and final Multilayer Perceptron (MLP) classification heads were randomly initialized and trained strictly from scratch.

5. Results

This section details the experimental evaluation of our proposed multimodal framework, structured around three primary diagnostic tasks. First, we establish the baseline efficacy of our architectures by benchmarking them on the English-language ADReSSo IS2021 challenge for AD classification. Following this, we extend our evaluation to the novel, Spanish-speaking SPIN cohort to tackle two complex clinical objectives: the binary prediction of $A\beta$ pathology and the challenging three-class differential diagnosis of Primary Progressive Aphasia (PPA) variants. Throughout the following subsections, we provide a comparative analysis between the baseline DMHAM model and the proposed CogniAlign architecture. Specifically, we examine the architectural impact of integrating handcrafted acoustic features (AF) and assess the models’ robust discriminative capabilities using standard classification metrics (Accuracy, macro-

Table 3: Results on the ADReSSo IS2021 challenge (binary: AD vs HC). AF = 55 handcrafted acoustic features. CogniAlign* = Results obtained from our independent replication of the architecture.

Model	Acc	Rec	Prec	F1
Luz et al. [10]	78.90	79.00	78.00	78.00
Syed et al. [41]	84.51	79.31	89.87	84.34
CogniAlign [17]	90.36	90.15	90.76	90.11
NeuroXVocal [39]	95.77	—	—	95.76
DMHAM	79.69	75.12	81.89	75.81
DMHAM + AF	79.34	77.17	78.67	77.23
CogniAlign*	87.20	87.32	87.14	87.10
CogniAlign* + AF	87.84	87.24	88.43	87.56

averaged F1-score, Recall, and Precision) alongside Receiver Operating Characteristic (ROC) curve analysis.

5.1. ADReSSo IS2021 Challenge

While the ADReSSo IS2021 dataset serves as our primary English-language benchmark, it is important to note that the objective of this study is not to exhaustively optimize for state-of-the-art performance on this specific challenge. Rather, we utilize ADReSSo to validate the structural soundness and baseline efficacy of the DMHAM and CogniAlign architectures on a well-documented standard. Establishing this foundational performance ensures that any subsequent findings on our primary target—the novel, multilingual, and clinically complex SPIN cohort—are grounded in a reliably validated architectural framework.

Table 3 presents the performance on the ADReSSo IS2021 challenge. Our re-implementation of the CogniAlign architecture achieves 87.20% accuracy. Although this is below the original reported performance (90.36% [17]), it remains highly competitive and successfully validates our deployment of the model. Furthermore, it significantly outperforms both the standard audio-only baseline [10] and DMHAM architecture (79.69%). While the addition of handcrafted Acoustic Features (AF) provides an improvement to the F1-score of the DMHAM architecture, overall accuracy decreased and yields only a marginal 0.7% relative improvement in accuracy for CogniAlign. We hypothesize that this plateau occurs because pause and silence metrics are among the most discriminative markers for the ADReSSo dataset. Because CogniAlign’s word-level cross-attention mechanism intrinsically encodes these temporal gaps and silences during multimodal alignment, the explicit injection of external pause-related features provides largely redundant information for this specific diagnostic task.

5.2. Amyloid-Beta Prediction (SPIN Cohort)

Table 4 presents the results for $A\beta$ prediction on the SPIN cohort, without transfer learning. This binary classification task is the primary clinical focus of this study, as predicting underlying $A\beta$ pathology non-invasively addresses a massive bottleneck in modern neuro-diagnostic pathways. Currently, confirming amyloid burden requires expensive Positron Emission Tomography (PET) scans or invasive lumbar punctures. Across all configurations, CogniAlign demonstrates highly robust, su-

Table 4: Results on amyloid-beta prediction (binary: positive vs negative)

Model	Acc	Rec	Prec	F1
DMHAM	60.22	56.46	62.32	52.12
DMHAM + AF	74.22	73.56	74.63	73.26
CogniAlign	79.70	78.70	79.24	78.90
CogniAlign + AF	83.86	83.19	84.33	83.44

perior performance in this challenging domain.

Without explicit acoustic features, CogniAlign establishes a strong baseline of 79.70% accuracy, vastly outperforming the DMHAM architecture (60.22% accuracy). More importantly, the addition of handcrafted acoustic features (AF) to CogniAlign pushes the predictive accuracy to a peak of 83.86% and achieves a high precision of 84.33%. In a clinical screening scenario, this high precision is paramount, as it directly minimizes false-positive referrals for costly confirmatory PET imaging.

A key finding here is the differential impact of acoustic features across tasks. Unlike the plateau observed in the symptomatic ADReSSo benchmark, adding AF to CogniAlign yields a substantial 5.2% relative accuracy improvement for early amyloid prediction. Given that CogniAlign already models temporal dynamics, this gain suggests that the presence of amyloid, even prior to severe cognitive decline, manifests in subtle micro-motor speech alterations. Markers within our 55-feature auxiliary set, such as fine-grained spectral variations and voice-quality phonatory metrics, successfully capture these deep, non-redundant biological signals. Furthermore, the best-performing configuration (CogniAlign + AF) yields an AUC of 0.803 ± 0.051 across the 5 cross-validation folds, confirming its strong and consistent discriminative capacity for ranking patient amyloid risk—a clinically critical property for pre-screening triage.

5.3. PPA 3-Class Classification (SPIN Cohort)

Table 5 details the performance on the three-way differential diagnosis of Primary Progressive Aphasia (PPA) variants. This multi-class task is inherently challenging due to subtle linguistic overlaps and severe class imbalances, with svPPA representing fewer than a third of the lvPPA samples. For reference, Themistocleous et al. [30] report 80% accuracy on a 3-class PPA task using a deep neural network on acoustic and linguistic features. Our best result of 73.40% accuracy is achieved by DMHAM+AF, closing much of this gap. The remaining difference is largely attributed to the substantially smaller and more imbalanced nature of our cohort, which introduced high variance across folds and made training inherently unstable—as noted in Section 4, this task carries higher uncertainty than the other tasks evaluated in this study.

Interestingly, the introduction of handcrafted Acoustic Features (AF) yields contrasting outcomes between the architectures. While AF provides a notable 19.7% relative F1-score boost to DMHAM, it degrades CogniAlign’s performance (dropping to 64.72% F1). A plausible explanation, consistent with the pattern observed in the ADReSSo and $A\beta$ tasks, is that CogniAlign’s Gated Cross-Attention may already capture much of the relevant prosodic and temporal information from the self-supervised embeddings, making the additional features partially redundant.

Table 5: Results on PPA 3-class classification (lvPPA / nvfPPA / svPPA)

Model	Acc	Rec	Prec	F1
DMHAM	67.97	51.60	47.80	48.72
DMHAM + AF	73.40	62.12	59.96	58.34
CogniAlign	73.30	67.60	70.21	66.60
CogniAlign + AF	71.34	67.02	65.53	64.72

6. Conclusions

In this study, we evaluated and adapted advanced multimodal deep learning frameworks to assess the diagnostic and predictive utility of spontaneous speech in neurodegenerative diseases. By benchmarking on the English ADReSSo dataset and extending our analysis to the Spanish/Catalan SPIN cohort, we demonstrated the viability of these approaches across different linguistic contexts. The top-performing architecture, CogniAlign, effectively integrates linguistic and acoustic modalities using word-level temporal alignment and Gated Cross-Attention, achieving highly competitive results across multiple clinical tasks.

Crucially, our findings reveal a highly task-dependent relationship regarding the utility of handcrafted acoustic features. For macroscopic clinical diagnostic tasks—specifically ADReSSo benchmarking and PPA sub-typing, CogniAlign’s self-supervised embeddings naturally captured sufficient low-level prosodic and spectral cues. In these instances, the addition of explicit acoustic features yielded either marginal improvements for AD classification or slight degradation for PPA sub-typing, rendering the complex extraction of external auxiliary features largely redundant for general phenotype classification.

However, the prediction of underlying $A\beta$ pathology presented a stark contrast. The explicit inclusion of handcrafted acoustic features provided a slight improvement to CogniAlign’s predictive accuracy on this task. This performance disparity suggests that while broad clinical syndromes (AD and PPA) can be reliably modeled via self-supervised latent spaces, the subtle, potentially early-stage physiological changes associated with $A\beta$ accumulation manifest in specific vocal biomarkers that still require explicit, targeted extraction.

Ultimately, this work highlights the tremendous potential of spontaneous speech as a non-invasive, scalable screening tool for both diagnostic classification and biomarker prediction. Future research will focus on exploring finer-grained multimodal integration, specifically transitioning from word-level to per-token audio-text alignment, to further refine the predictive capabilities of these models.

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